

HALLOCK, MN, USA

WMO: 727478

Lat: 48.753N

Lon: 96.943W

Elev: 250

StdP: 98.36

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04934

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.2	-27.9	-35.0	0.1	-31.4	-32.2	0.2	-27.9	14.4	-12.7	13.2	-12.2	2.7	150	0.683

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	31.4	21.8	29.1	20.7	27.8	20.0	24.5	28.7	23.0	27.3	21.8	26.1	5.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.2	26.6	22.1	17.3	26.0	20.5	15.6	24.5	75.4	29.0	69.6	26.9	64.9	25.7	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.3	10.1	DB	-33.7	34.8	3.4	1.5	-36.2	35.9	-38.2	36.8	-40.1	37.7	-42.6	38.8	(4)
(5)			WB	-33.8	26.5	3.3	1.4	-36.2	27.5	-38.1	28.3	-40.0	29.1	-42.4	30.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-14.5	-12.9	-4.8	5.1	12.1	17.8	20.1	19.2	14.5	6.5	-2.6	-10.5	(6)	
	DBStd	13.67	8.16	7.86	7.75	5.72	5.02	3.72	3.12	3.40	4.78	5.16	6.38	7.25	(7)	
	HDD10.0	3221	759	642	459	163	36	1	0	0	13	133	379	636	(8)	
	HDD18.3	5327	1017	876	717	396	201	52	18	31	132	367	629	894	(9)	
	CDD10.0	1127	0	0	1	17	101	237	312	284	149	26	1	0	(10)	
	CDD18.3	192	0	0	0	0	8	37	72	56	18	1	0	0	(11)	
	CDH23.3	2241	0	0	0	10	147	418	752	664	234	16	0	0	(12)	
	CDH26.7	666	0	0	0	1	43	116	223	212	68	2	0	0	(13)	
(14) Wind	WSAvg	4.5	4.7	4.7	4.8	5.0	5.1	4.2	3.5	3.6	4.4	4.8	4.8	4.6	(14)	
(15) Precipitation	PrecAvg	479	17	12	21	33	60	81	71	58	58	35	18	16	(15)	
	PrecMax	744	41	40	71	76	157	146	185	193	154	100	58	37	(16)	
	PrecMin	287	2	0	0	0	9	17	16	12	5	3	0	1	(17)	
	PrecStd	104	10	9	15	23	35	34	40	39	39	27	14	8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	6.3	17.3	24.6	31.3	32.5	32.9	33.8	32.0	26.0	14.8	4.1	(19)	
		MCWB	1.6	4.7	12.8	13.2	17.5	21.6	24.5	22.8	20.1	16.1	9.9	2.1	(20)	
	2%	DB	1.3	2.8	11.0	21.2	27.4	29.1	31.1	31.2	28.6	21.2	11.2	2.4	(21)	
		MCWB	0.6	1.8	7.8	11.4	15.3	19.9	23.3	21.9	19.5	14.5	7.9	1.3	(22)	
	5%	DB	-1.2	1.3	7.3	17.8	24.0	27.6	28.9	28.8	26.2	17.6	7.9	1.1	(23)	
		MCWB	-1.5	0.6	5.2	9.8	13.9	19.1	22.3	20.5	17.5	12.3	5.7	0.2	(24)	
	10%	DB	-2.8	-1.2	4.0	15.1	22.1	26.1	27.5	27.3	23.0	14.8	5.9	-1.1	(25)	
		MCWB	-3.5	-1.6	2.6	8.6	13.3	18.4	21.3	19.8	16.6	11.0	3.8	-1.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	5.0	13.8	15.0	19.8	25.4	27.1	25.4	23.1	17.7	11.0	2.8	(27)	
		MCDB	3.1	6.7	15.9	20.9	24.6	28.7	30.2	30.2	28.3	22.1	13.9	3.7	(28)	
	2%	WB	0.5	2.2	7.8	12.4	17.9	22.7	25.1	23.6	20.8	15.4	8.1	1.3	(29)	
		MCDB	1.4	2.9	9.9	18.6	22.5	27.0	29.1	28.6	26.0	20.0	10.0	2.2	(30)	
	5%	WB	-1.8	0.5	5.0	10.8	16.3	20.9	23.6	22.3	19.1	13.1	6.0	-0.1	(31)	
		MCDB	-1.6	1.3	7.2	16.4	21.0	24.9	27.4	26.8	23.7	17.2	8.3	0.8	(32)	
	10%	WB	-3.7	-2.0	3.0	9.2	14.8	19.6	22.3	20.9	17.6	11.0	3.8	-1.5	(33)	
		MCDB	-2.8	-1.6	4.0	14.7	20.2	23.9	26.2	25.0	21.7	14.4	5.6	-1.2	(34)	
(35) Mean Daily Temperature Range	MDBR		9.4	10.3	9.6	11.6	13.0	11.9	12.6	13.7	13.1	10.1	8.5	8.4	(35)	
	5% DB	MCDBR	9.4	9.1	11.2	17.4	17.4	15.1	14.7	16.2	16.6	15.1	11.9	9.7	(36)	
		MCWBR	8.8	8.0	8.2	9.8	8.6	7.6	8.4	8.1	8.6	9.1	8.8	8.6	(37)	
	5% WB	MCDBR	9.3	8.8	10.8	15.2	13.2	12.2	13.0	13.5	13.7	13.4	11.2	9.2	(38)	
		MCWBR	8.8	7.8	8.1	9.4	7.9	7.4	8.5	8.0	8.1	8.5	8.5	8.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.281	0.305	0.305	0.357	0.373	0.388	0.390	0.386	0.352	0.310	0.280	0.267	(40)	
	taud		2.241	2.184	2.390	2.339	2.357	2.341	2.344	2.356	2.446	2.532	2.490	2.362	(41)	
	Ebn at Noon		771	834	913	891	887	873	865	851	851	841	793	754	(42)	
	Edh at Noon		75	101	98	115	118	121	119	112	93	72	59	59	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.50	2.74	3.94	4.89	5.48	5.89	6.29	5.20	3.80	2.31	1.32	1.03	(44)	
	RadStd		0.16	0.19	0.38	0.43	0.38	0.45	0.27	0.34	0.27	0.32	0.17	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page